Unit-5: Who Decides What Governance the Organization Needs?

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So, Who Are the Gatekeepers?

SHAREHOLDERS
(Driven by Value & Sustainability)

REGULATORS
(Driven by Safety & Law)

Trusted by public,approved by regulators,valuable to shareholders

THE PUBLIC
(Driven by Trust & Fairness)

Governance is not a barrier to innovation. It's the blueprint for building trustworthy AI that succeeds in the real world.

Gatekeepers of governance

- **The Current Scenario**
  - Stakeholders are demanding two things:
    Return on Investment (**ROI**)
    Assurance against future risks.
  - Formal regulations (like national laws) are often **slow to identify risks** and reflect a **historical understanding** of problems.

- **The Real Power: Public Opinion**
  - Public opinion acts as the ultimate gatekeeper because it determines:
    - **Where** people invest money
    - **Which** products they buy.
    - **What** rules governments eventually make

When Trust Fails: Lessons from History

- **Trust vs. Tech**
  - Case: Agricultural biotechnology in the 1990s,.
  - Outcome: Despite scientific arguments about safety, public opinion in Europe swung against them.

- **The "Malicious" Lesson**
  - Case: Facebook-Cambridge Analytica.
  - Outcome: Showed how ML can be used with explicitly malicious intent to manipulate opinion.

- **The "Unintentional" Lesson**
  - Case: Recruitment tools and criminal assessment systems.
  - Outcome: "Black box" judgments resulted in illegal bias (race/gender) without the engineers intending to do harm

Matching Governance with Risk Level

There are several dimensions to consider when assessing risk, including:

- The audience for the model
- The lifetime of the model and its outcomes
- The impact of the outcomes

The type of KPIs chosen by the business, the type of model building algorithm used for the required level of explainability, the coding tools used, the level of documentation and reproducibility, the level of automated testing, the resilience of the hardware platform, and the type of monitoring implemented

Choosing the right kind of operationalization model and MLOps features depending on the project's criticality

Project criticality	Operationalization	Builder autonomy	Versioning	Resources separation	SLA and support by IT	Integration to ext. systems
Irregular Ad-hoc usage	SSA with run on design node	★★★	—	—	—	—
Scheduled but can be inoperative for a small amount of time	Self-service development and scheduling	★★★	★★★	★★	—	—
Scheduled and requires specific monitoring	Light deployment process with rough QA and scheduling	★	★★★	★★★	★	—
Operational projects that cannot suffer outages	Fully controlled deployment CI/CD	—	★★★	★★★	★★★	★★★

Solution: Responsible AI

"What is profitable?" => "What is trustworthy?".

- Defining the Standard
  - Responsible AI is the modern consensus that ML systems must be Accountable, Sustainable, and Governable.
  - Because of this framework, we must implement specific controls in our pipelines:
    - Data Provenance: To ensure we know where data comes from.
    - Bias Detection: To stop the "Black Box" from discriminating.
    - Explainability: To prove to regulators how the decision was made

Recap:

- Public opinion acts as the ultimate and fastest-moving gatekeeper for governance, affecting investment and regulation
- Successful governance requires ML systems to be trusted by the public, approved by regulators, and valuable to shareholders
- Loss of public trust, caused by malicious use or unintentional bias, prevents technological benefits and leads to negative outcomes
- Governance levels are matched with the application scope
- Responsible AI is the modern framework (Accountable, Sustainable, Governable) that shifts the engineering goal to trustworthiness and requires technical controls like bias detection

Thank You!